CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

ORDER NO. 98-211

WASTE DISCHARGE REQUIREMENTS
FOR
CLEARLAKE OAKS COUNTY WATER DISTRICT
WASTEWATER TREATMENT PLANT
LAKE COUNTY

The California Regional Water Quality Control Board, Central Valley Region, (hereafter Board) finds that:

1. Clearlake Oaks County Water District (hereafter Discharger) owns and operates a wastewater collection, treatment and disposal system that serves the community of Clearlake Oaks on the east side of Clear Lake in Lake County.

2. The plant is in Section 32, T14N, R7W, MDB&M, with surface water drainage to Clear Lake, as shown in Attachment A, which is attached hereto and part of the Order by reference.

3. On 28 January 1994, the Board adopted Waste Discharge Requirements (WDRs) No. 94-028 which prescribes requirements for the treatment of approximately 0.43 million gallons per day (mgd) of domestic wastewater in an oxidation ditch and clarifier, with discharge to disposal ponds. The maximum daily discharge and monthly average dry weather discharge design capacities are 2.1 and 0.5 mgd, respectively.

4. The Discharger has received concept approval for small community Grant Program funding which will enable it to begin construction for pumping treated effluent to the Lake County Sanitation District's (LACOSAN) Southeast Regional Wastewater Treatment (SERWT) Facility. This project includes treatment plant upgrades, collection system rehabilitation, emergency storage basin enhancements, and installation of pipeline to the SERWT facility. A general facility map is as shown in Attachment B, and a general alignment of the effluent pipeline is as shown in Attachment C, which are attached hereto and part of the Order by reference.

5. Order No. 94-028 is neither adequate nor consistent with current plans and policies of the Board.

6. The Board adopted a Water Quality Control Plan, Fourth Edition, for the Sacramento River Basin and San Joaquin River Basin (hereafter Basin Plan), which contains water quality objectives. These requirements are consistent with that Plan.

7. In the past, the disposal ponds have relied on percolation and evaporation to dispose of the treated wastewater. During the winter months, the groundwater rises and severely inhibits percolation of the ponds, thus reducing the disposal capacity. These requirements include a monitoring program to evaluate any impacts to groundwater. Additionally, the sewage collection system is inundated when the groundwater rises and this contributes some infiltration to the collection system. The infiltration coupled with inflow to the system triples the influent to the plant during wet periods with high lake levels. The inflow and infiltration (I/I) is more than the disposal system capacity during high groundwater periods. The Discharger is currently under a Cease and Desist Order No. 94-029 adopted 28 January 1994 requiring the District to evaluate discharge options and construct adequate treatment and disposal facilities. Cease and Desist Order No. 94-029 shall remain in full force and effect until rescinded.

8. The discharges to Clear Lake are violations of a prohibition contained in the Basin Plan.

9. The beneficial uses of Clear Lake are municipal, industrial, and agricultural supply; recreation; aesthetic enjoyment; navigation; groundwater recharge; fresh water replenishment; hydropower generation; and preservation and enhancement of fish, wildlife, and other aquatic resources.

10. The beneficial uses of underlying groundwater are domestic, agricultural, and industrial supply.

11. The action to adopt waste discharge requirements for this facility is exempt from the provisions of the California Environmental Quality Act (CEQA), in accordance with Section 15301(b), Title 14, California Code of Regulations (CCR).

12. This discharge is exempt from the requirements of Consolidated Regulations for Treatment, Storage, Processing, or Disposal of Solid Waste, as set forth in Title 27, CCR, Division 2, Subdivision 1, Section 20005, et seq., (hereafter Title 27). The exemption, pursuant to Section 20090(b), is based on the following:

    a. The Board is issuing waste discharge requirements,

    b. The permitted discharge complies with the Basin Plan, and

    c. The wastewater does not need to be managed according to Title 22, CCR, Division 4, Chapter 30, as a hazardous waste.

13. The Discharger and LACOSAN entered into an "Agreement for the Acceptance of Effluent from the Clearlake Oaks County Water District," dated 21 July 1998. This agreement provides the Discharger with the terms and conditions to allow the discharge of treated effluent to LACOSAN's Southeast Reservoir. This agreement also outlines conditions including, but not limited to, the term of the agreement (twenty years), operation and maintenance responsibilities, delivery, a normal flow rate of 1500 gpm, and operation emergencies, repair and reconstruction responsibilities. The agreement states that the Discharger shall comply with the requirements of the Operating Agreement Southeast Geysers Effluent Pipeline Project (Operating Agreement) between LACOSAN, Calpine Corporation, Northern California Power Agency (NCPA), and UNOCAL or their successors. These WDRs address the collection, treatment, and discharge of the treated effluent to the Southeast Reservoir for final reuse via the Geyser Effluent Pipeline Project.

14. The Board has notified the Discharger and interested agencies and persons of its intent to prescribe waste discharge requirements for this discharge.

15. The Board, in a public meeting, heard and considered all comments pertaining to the discharge.

IT IS HEREBY ORDERED that Order No. 94-028 is rescinded and Clearlake Oaks County Water District, its agents, successors, and assigns, in order to meet the provisions contained in Division 7 of the California Water Code and regulations adopted thereunder, shall comply with the following:

A. Discharge Prohibitions:

1. Discharge of wastes to surface waters or surface water drainage courses is prohibited.

2. Bypass or overflow of untreated or partially treated waste is prohibited.

3. Discharge of waste classified as 'hazardous' or 'designated', as defined in Sections 2521(a) and 2522(a) of Chapter 15, is prohibited.

4. Discharge or release of effluent from any segment of the pipeline, other than at the delivery point, is prohibited.

B. Discharge Specifications:

1. The monthly average dry weather (design) discharge flow shall not exceed 0.5 mgd.

2. The maximum daily discharge shall not exceed 2.1 million gallons.

3. Objectionable odors originating at this facility shall not be perceivable beyond the limits of property owned by the Discharger.

4. As a means of determining compliance with Discharge Specification No. 3, the dissolved oxygen content in the upper zone (1 foot) of wastewater in ponds shall not be less than 1.0 mg/l.

5. The treatment facilities shall be designed, constructed, operated, and maintained to prevent inundation or washout due to floods with a 100-year return frequency.

6. The effluent from the treatment facility to the pipeline and storage ponds shall not exceed the following limits:

Constituent	Units	30-day Average	Monthly Median	Monthly Maximum
BOD_5[1]	mg/l	40	---	80
Settleable Solids	ml/l	0.2	---	0.5
Total Coliform	MPN/100 mL	---	23	230

[1] Five-day, 20° Celsius biochemical oxygen demand.

7. Ponds shall not have a pH less than 6.5 or greater than 8.5.

8. Ponds shall be managed to prevent breeding of mosquitoes. In particular,

a. An erosion control program should assure that small coves and irregularities are not created around the perimeter of the water surface.

b. Weeds shall be minimized through control of water depth, harvesting, or herbicides.

c. Dead algae, vegetation, and debris shall not accumulate on the water surface.

9. Public contact with wastewater shall be precluded through such means as fences, signs, and other acceptable alternatives.

10. Ponds shall have sufficient capacity to accommodate allowable wastewater flow, design seasonal precipitation, and ancillary inflow and infiltration. Design seasonal precipitation shall be based on total annual precipitation using a return period of 100 years, distributed monthly in accordance with historical rainfall patterns. Freeboard shall never be less than two feet (measured vertically to the lowest point of overflow).

11. On or about **1 October** of each year, available pond storage capacity shall at least equal the volume necessary to comply with Discharge Specification No. B.10.

12. Emergency storage basins shall not contain treated or untreated wastewater for more than 20 days.

C. Sludge Disposal:

1. Collected screenings, sludges, and other solids removed from liquid wastes shall be disposed of in a manner that is consistent with Title 27 CCR, Division 2, Subdivision 1, Section 20005, et seq., and approved by the Executive Officer.

2. Any proposed change in sludge use or disposal practice from a previously approved practice shall be reported to the Executive Officer and U.S. Environmental Protection Agency (EPA) Regional Administrator at least **90 days** in advance of the change.

3. Use and disposal of sewage shall comply with existing Federal and State laws and regulations, including permitting requirements and technical standards included in 40 CFR 503.

   If the State Water Resources Control Board and the Regional Water Quality Control Boards are given the authority to implement regulations contained in 40 CFR 503, this Order may be reopened to incorporate appropriate time schedules and technical standards. The Discharger must comply with the standards and time schedules contained in 40 CFR 503 whether or not they have been incorporated into this Order.

4. The Discharger is encouraged to comply with the State Guidance Manual issued by the Department of Health Services titled *Manual of Good Practice for Landspreading of Sewage Sludge.*

5. The Discharger shall submit annually a sludge disposal plan describing the annual volume of sludge generated by the plant and specifying the disposal practices.

D. Groundwater Limitations:

The discharge shall not cause underlying groundwater to be degraded.

E. Provisions:

1. The Discharger shall comply with the Monitoring and Reporting Program No. 98-211, which is part of this Order, and any revisions thereto as ordered by the Executive Officer.

2. The Discharger shall comply with the "Standard Provisions and Reporting Requirements for Waste Discharge Requirements," dated 1 March 1991, which are attached hereto and by reference a part of this Order. This attachment and its individual paragraphs are commonly referenced as "Standard Provision(s)."

3. The Discharger shall comply with the following time schedule:

	Task	Completion Date
a.	Submit finalized agreement providing the facility access to the geysers pipeline	**15 November 1998**
b.	Submit Groundwater Assessment Plan	**1 February 1999**
c.	Submit Surface Impoundments and Emergency Storage Basins Monitoring Plan	**1 February 1999**
d.	Complete construction of approved groundwater monitoring facilities	**1 May 1999**
e.	Complete connection to geyser pipeline	**1 December 1999**
f.	Submit Groundwater Assessment Report	**1 November 1999**

4. In the event of any change in control or ownership of land or waste discharge facilities described herein, the Discharger shall notify the succeeding owner or operator of the existence of this Order by letter, a copy of which shall be immediately forwarded to this office.

5. At least **90 days** prior to termination or expiration of any lease, contract, or agreement involving disposal or reclamation areas or off-site reuse of effluent, used to justify the capacity authorized herein and assure compliance with this Order, the Discharger shall notify the Board in writing of the situation and of what measures have been taken or are being taken to assure full compliance with this Order.

6. The Discharger must comply with all conditions of this Order, including timely submittal of technical and monitoring reports as directed by the Executive Officer. Violations may result in enforcement action, including Board or court orders requiring corrective action or imposing civil monetary liability, or in revision or rescission of this Order.

7. A copy of this Order shall be kept at the discharge facility for reference by operating personnel. Key operating personnel shall be familiar with its contents.

8. If reclaimed water is used for construction purposes, it shall comply with the most current edition of "Guidelines for Use of Reclaimed Water for Construction Purposes". Other uses of reclaimed water not specifically authorized herein shall be subject to the approval of the Executive Officer and shall comply with Title 22, CCR, Division 4.

9. The Board will review this Order periodically and will revise requirements when necessary.

I, GARY M. CARLTON, Executive Officer, do hereby certify the foregoing is a full, true, and correct copy of an Order adopted by the California Regional Water Quality Control Board, Central Valley Region, on 23 October 1998.

GARY M. CARLTON, Executive Officer

AMENDED
wac/lsb:clrlakeo.wdr

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

REVISED MONITORING AND REPORTING PROGRAM NO. 98-211

FOR
CLEARLAKE OAKS COUNTY WATER DISTRICT
WASTEWATER TREATMENT PLANT
LAKE COUNTY

This monitoring and reporting program (MRP) incorporates requirements for monitoring of the wastewater treatment plant. This MRP is issued pursuant to Water Code Section 13267. The Discharger shall not implement any changes to this MRP unless and until a revised MRP is issued by the Executive Officer.

All wastewater samples should be representative of the volume and nature of the discharge. The time, date, and location of each grab sample shall be recorded on the sample chain of custody form. Process wastewater flow monitoring shall be conducted continuously using a flow meter and shall be reported in cumulative gallons per day.

Field test instruments (such as pH and dissolved oxygen) may be used provided that:

1. The operator is trained in the proper use of the instrument;
2. The instruments are field calibrated at the frequency recommended by the instrument manufacturer or at the frequency recommended by an industry standard;
3. Instruments are serviced and/or calibrated by the manufacturer at the recommended frequency; and
4. Field calibration reports are submitted as described in the "Reporting" section of this MRP.

INFLUENT MONITORING

Constituent	Units	Type of Sample	Sampling Frequency	Reporting Frequency
Flow	gpd	Continuous	Daily	Monthly
BOD_5 [1]	mg/L	Grab	Monthly	Monthly

[1] 5-day, 20°C Biochemical Oxygen Demand.

EFFLUENT MONITORING

Effluent samples shall be collected at the effluent pump station or from the pipeline that discharges to the emergency storage ponds. Effluent monitoring shall include at least the following:

Constituent	Units	Type of Sample	Sampling Frequency	Reporting Frequency
Total Coliform Organisms [1]	MPN/100 mL [2]	Grab	Weekly	Monthly

Constituent	Units	Type of Sample	Sampling Frequency	Reporting Frequency
BOD_5 [3]	mg/L	Grab	Twice Monthly	Monthly
Settleable Solids	mg/L	Grab	Monthly	Monthly
Total Dissolved Solids	mg/L	Grab	Monthly	Monthly
Nitrate as Nitrogen	mg/L	Grab	Monthly	Monthly
Total Kjeldahl Nitrogen	mg/L	Grab	Monthly	Monthly
Specific Conductivity@ 25°C	µmhos/cm	Grab	Monthly	Monthly
Standard Minerals[4]	mg/L	Grab	Annually	Annually

1 Using a minimum of 10 tubes or two dilutions.
2 Most probable number per 100 ml.
3 5-day, 20°C Biochemical Oxygen Demand.
4 Standard Minerals shall include the following: boron, calcium, iron, manganese, magnesium, potassium, sodium, chloride, total alkalinity (including alkalinity series), and hardness.

POND MONITORING

Each of the wastewater treatment ponds shall be monitored for the parameters specified below. If the emergency storage ponds are used for the temporary storage of wastewater, then the ponds shall be monitored for the parameters listed below. In addition, the Discharger shall submit a written report to the Regional Board within ten days that includes the reason and period of use, volume stored, duration of storage, and water quality measurements taken.

Constituent	Units	Type of Sample	Sampling Frequency	Reporting Frequency
Dissolved Oxygen[1]	mg/L	Grab	Weekly	Monthly
pH	pH units	Grab	Weekly	Monthly
Freeboard	0.25 feet	Observation	Weekly	Monthly
Berm Seepage[2]	NA	Observation	Weekly	Monthly
Odors[3]	--	Observation	Weekly	Monthly

1 Samples shall be collected at a depth of one foot from each pond in use, opposite the inlet. Samples shall be collected between 0700 and 0900 hours.
2 Reservoir containment levees shall be observed for signs of seepage or surfacing water along the exterior toe of the levees and dam. If surfacing water is found, then a sample shall be collected and tested for total coliform organisms and total dissolved solids.
3 The presence of strong or unusual odors shall be reported.

SLUDGE MONITORING

A composite sample of sludge shall be collected prior to sludge removal from the disposal beds in accordance with EPA's POTW Sludge Sampling and Analysis Guidance Document, August 1989, and tested for the following metals:

Arsenic	Cadmium	Copper	Chromium
Mercury	Molybdenum	Lead	Nickel
Selenium	Zinc		

Sampling records shall be retained for a minimum of five years. A log shall be kept of sludge quantities generated and of handling and disposal activities. The frequency of entries is discretionary; however, the log should be complete enough to serve as a basis for part of the annual report.

WATER SUPPLY MONITORING

A sampling station within the distribution system shall be utilized to collect a representative sample of the municipal water supply can be obtained. Water supply monitoring shall include at least the following constituents. As an alternative to annual water supply monitoring, the Discharger may submit results of the most current DHS water supply monitoring.

Constituents	Units	Sampling Frequency	Reporting Frequency
Specific Conductivity@ 25°C	µmhos/cm	Annually	Annually
pH	pH units	Annually	Annually
Nitrate as Nitrogen	mg/L	Annually	Annually

GROUNDWATER MONITORING

The groundwater monitoring program shall begin in the third quarter of 2004. Prior to construction and/or sampling of any groundwater monitoring wells, the Discharger shall submit plans and specifications to the Regional Board for review and approval. Once installed, all new wells shall be added to the MRP and shall be sampled and analyzed according to the schedule below.

Prior to sampling, the groundwater elevations shall be measured and the wells shall be purged at least three well volumes until temperature, pH and electrical conductivity have stabilized. Depth to groundwater shall be measured to the nearest 0.01 feet. Samples shall be collected using standard EPA methods. Groundwater monitoring shall include, at a minimum, the following:

Constituents	Units	Type of Sample	Sampling Frequency	Reporting Frequency
Depth to Groundwater	0.01 feet	Measurement	Quarterly	Quarterly
Groundwater Elevation[1]	0.01 feet	Calculated	Quarterly	Quarterly

Gradient	feet/feet	Calculated	Quarterly	Quarterly
Gradient Direction	Degrees	Calculated	Quarterly	Quarterly
pH[2]	pH units	Grab	Quarterly	Quarterly
Specific Conductivity@ 25°C	µmhos/cm	Grab	Quarterly	Quarterly
Total Coliform Organisms	MPN/100 mL	Grab	Quarterly	Quarterly
Total Dissolved Solids	mg/L	Grab	Quarterly	Quarterly
Nitrate as Nitrogen	mg/L	Grab	Quarterly	Quarterly
Total Kjeldahl Nitrogen	mg/L	Grab	Quarterly	Quarterly
Standard Minerals[3]	mg/L	Grab	Annually	Annually

[1] Groundwater elevation shall be determined based on depth-to-water measurements from a surveyed measuring point elevation on the well.
[2] Hand held field meter may be used.
[3] Standard Minerals shall include the following: boron, calcium, iron, manganese, magnesium, potassium, sodium, chloride, total alkalinity (including alkalinity series), and hardness.

REPORTING

In reporting monitoring data, the Discharger shall arrange the data in tabular form so that the date, sample type (e.g., process wastewater effluent, groundwater well, etc.), and reported analytical result for each sample are readily discernible. The data shall be summarized in such a manner to clearly illustrate compliance with waste discharge requirements and spatial or temporal trends, as applicable. The results of any monitoring done more frequently than required at the locations specified in the Monitoring and Reporting Program shall be reported in the next scheduled monitoring report.

As required by the California Business and Professions Code Sections 6735, 7835, and 7835.1, all Quarterly Groundwater Monitoring Reports and the annual groundwater evaluation shall be prepared under the direct supervision of a Registered Engineer or Geologist and signed by the registered professional.

A. Monthly Monitoring Reports

Monthly reports shall be submitted to the Regional Board by the 1st **day of the second month** following the end of the reporting period (i.e. the August monthly report is due by 1 October). Monthly reports for the months of March, June, September, and December may be submitted as part of the Quarterly Monitoring Report, if desired. The monthly reports shall include the following:

1. Results of influent, effluent, and pond monitoring;

2. A comparison of monitoring data to the discharge specifications and an explanation of any violation of those requirements. Data shall be presented in tabular format;

3. If requested by staff, copies of laboratory analytical report(s); and

4. A calibration log verifying calibration of all hand held monitoring instruments and devices used to comply with the prescribed monitoring program;

B. Quarterly Monitoring Reports

Beginning with the third quarter of 2004, the Discharger shall establish a quarterly sampling schedule for groundwater monitoring such that samples are obtained approximately every three months. Quarterly monitoring reports shall be submitted to the Regional Board by the 1st **day of the second month after the quarter** (i.e. the January-March quarterly report is due by May 1st) and may be combined with the monthly report. The Quarterly Report shall include the following:

1. Results of groundwater monitoring;

2. A narrative description of all preparatory, monitoring, sampling, and analytical testing activities. The narrative shall be sufficiently detailed to verify compliance with the WDRs, this MRP, and the Standard Provisions and Reporting Requirements. The narrative shall be supported by field logs for each well documenting depth to groundwater; parameters measured before, during, and after purging; method of purging; calculation of the casing volume; and total volume of water purged;

3. Calculation of groundwater elevations, an assessment of the groundwater flow direction and gradient on the date of measurement, comparison to previous flow direction and gradient data, and discussion of seasonal trends, if any;

4. A narrative discussion of the analytical results for all media and locations monitored, including spatial and temporal trends, with reference to summary data tables, graphs, and appended analytical reports (as applicable);

5. A comparison of monitoring data to the discharge specifications, groundwater limitations, and surface water limitations, and explanation of any violation of those requirements;

6. Summary data tables of historical and current water table elevations and analytical results;

7. A scaled map showing relevant structures and features of the facility, the locations of monitoring wells and other sampling stations, and groundwater elevation contours referenced to mean sea level datum; and

8. Copies of laboratory analytical report(s) for groundwater monitoring.

C. Annual Monitoring Reports

An Annual Report shall be prepared as the fourth quarter monitoring report and shall include all monitoring data required in the monthly/quarterly schedule. The Annual Report shall be submitted to the Regional Board by **1 February of each year** and shall include the following:

1. The contents of the regular groundwater monitoring report for the last sampling event of the year;
2. If requested by staff, tabular and graphical summaries of all data collected during the year;
3. Data for monitoring the effluent, groundwater, and supply water performed on an annual basis;
4. An evaluation of the groundwater quality beneath the facility;
5. An evaluation of the performance of the wastewater treatment system, as well as a forecast of the flows anticipated in the next year;
6. Verification of appropriate employee training for all personnel involved in operation and maintenance of wastewater treatment system;
7. A discussion of compliance and the corrective action taken, as well as any planned or proposed actions needed to bring the discharge into full compliance with the waste discharge requirements; and
8. A discussion of any data gaps and potential deficiencies/redundancies in the monitoring system or reporting program.

A letter transmitting the self-monitoring reports shall accompany each report. Such a letter shall include a discussion of requirement violations found during the reporting period, and actions taken or planned for correcting noted violations, such as operation or facility modifications. If the discharger has previously submitted a report describing corrective actions and/or a time schedule for implementing the corrective actions, reference to the previous correspondence will be satisfactory. The transmittal letter shall contain a statement by the discharger, or the discharger's authorized agent, under penalty of perjury, that to the best of the signer's knowledge the report is true, accurate and complete.

The Discharger shall implement the above monitoring program as of the date of this Order.

Ordered by: Thomas R Pinkos

THOMAS R. PINKOS, Executive Officer

4 November 2003

(Date)

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

MONITORING AND REPORTING PROGRAM NO. 98-211

FOR
CLEARLAKE OAKS COUNTY WATER DISTRICT
WASTEWATER TREATMENT PLANT
LAKE COUNTY

Sample stations shall be established such that the samples are representative of the volume and nature of the discharge. Specific sample station locations shall be established under direction of the Board's staff and a description of the stations shall be attached to this Order.

EFFLUENT MONITORING

Effluent samples shall be collected just prior to discharge to the disposal facility. Effluent samples should be representative of the volume and nature of the discharge. Samples collected from the outlet structure of ponds will be considered adequately composited. Time of collection of a grab sample shall be recorded. Effluent monitoring shall include at least the following:

Constituents	Units	Type of Sample	Sampling Frequency
Flow	mgd	Cumulative	Daily
BOD_5 @ 20°C	mg/l	Grab	Monthly
Specific Conductivity	µmhos/cm	Grab	Monthly
pH	pH Units	Grab	Monthly
Nitrates as N	mg/l	Grab	Quarterly
Standard Minerals	mg/l	Grab	Annually

SLUDGE MONITORING

A composite sample of sludge shall be collected when sludge is removed from the disposal ponds, or annually when sludge is wasted from the clarifier, in accordance with EPA's POTW Sludge Sampling and Analysis Guidance Document, August 1989, and tested for the following metals:

Arsenic	Cadmium	Copper	Chromium
Mercury	Molybdenum	Lead	Nickel
Selenium	Zinc		

Sampling records shall be retained for a minimum of five years. A log shall be kept of sludge quantities generated and of handling and disposal activities. The frequency of entries is discretionary; however, the log should be complete enough to serve as a basis for part of the annual report.

WATER SUPPLY MONITORING

A sampling station shall be established where a representative sample of the municipal water supply can be obtained. Water supply monitoring shall include at least the following:

Constituents	Units	Sampling Frequency
pH	pH units	Annually
Electrical Conductivity @ 25°C	µmhos/cm	Annually
Standard Minerals	mg/l	Annually

GROUNDWATER MONITORING

Prior to construction, plans and specifications for groundwater monitoring wells shall be submitted to Board staff for review and approval. The approved monitoring wells shall be sampled and tested as follows:

Constituents	Units	Sampling Frequency
Groundwater Elevation	feet, MSL (± 0.01)	As Noted[1]
pH	pH units	Quarterly
Electrical Conductivity @25°C	µmhos/cm	Quarterly
Total Dissolved Solids	mg/l	Quarterly
Total Coliform	MPN/100 ml	Quarterly
Standard Minerals	mg/l	Annually

[1] Monthly November through May, July and September

UNAUTHORIZED SURFACE WATER DISCHARGES

During periods of unauthorized discharges to Clear Lake, the Discharger is required to sample the discharge and receiving water in accordance with the following table:

Effluent Monitoring:

Constituents	Units	Sampling Frequency
Flow (measured)	mgd	Daily
pH	pH units	Daily
Specific Conductivity	µmhos/cm	Daily
Total Coliform	MPN/100 ml	Daily
Total Suspended Solids	mg/l	Weekly
BOD_5[1]	mg/l	Weekly

[1] Biochemical Oxygen Demand, 20°C, 5-Day

Receiving Water Monitoring:

Specific sample locations shall be established. All constituents shall be sampled at each sample location.

Constituents	Units	Sampling Frequency
pH	pH units	Weekly
Specific Conductivity	µmhos/cm	Weekly
Total Coliform	MPN/100 ml	Weekly
Dissolved Oxygen	mg/l	Weekly
Turbidity	NTU	Weekly

In conducting the receiving water sampling, a log shall be kept of the receiving water conditions throughout the area nearest the discharge point. Attention shall be given to the presence or absence of:

a. Floating or suspended matter
b. Discoloration
c. Bottom deposits
d. Aquatic life
e. Fungi, slimes, or objectionable growths
f. Potential nuisance conditions
g. Visible films, sheens or coatings

Notes on receiving water conditions shall be summarized in the monitoring report.

EMERGENCY STORAGE BASIN MONITORING

The level of effluent in each emergency storage basin shall be measured to the nearest 0.1 feet of freeboard and reported monthly. The use of each basin shall be recorded and reported to the Board in writing within ten days. This report shall include the reason and period of use, volume stored, duration of storage and any water quality measurements taken. Monthly monitoring reports shall discuss all emergency storage event(s) occurring during the month.

REPORTING

In reporting the monitoring data, the Discharger shall arrange the data in tabular form so that the date, the constituents, and the concentrations are readily discernible. The data shall be summarized in such a manner to illustrate clearly the compliance with waste discharge requirements.

Monthly monitoring reports shall be submitted to the Regional Board by the **15th day** of the following month.

The results of any monitoring done more frequently than required at the locations specified in the Monitoring and Reporting Program shall be reported to the Board.

The Discharger shall submit an annual report to the Board by **30 January** of each year. The report shall contain both tabular and graphical summaries of the monitoring data obtained during the previous year. In addition, the Discharger shall discuss the compliance record and the corrective actions taken or planned which may be needed to bring the discharge into full compliance with the waste discharge requirements.

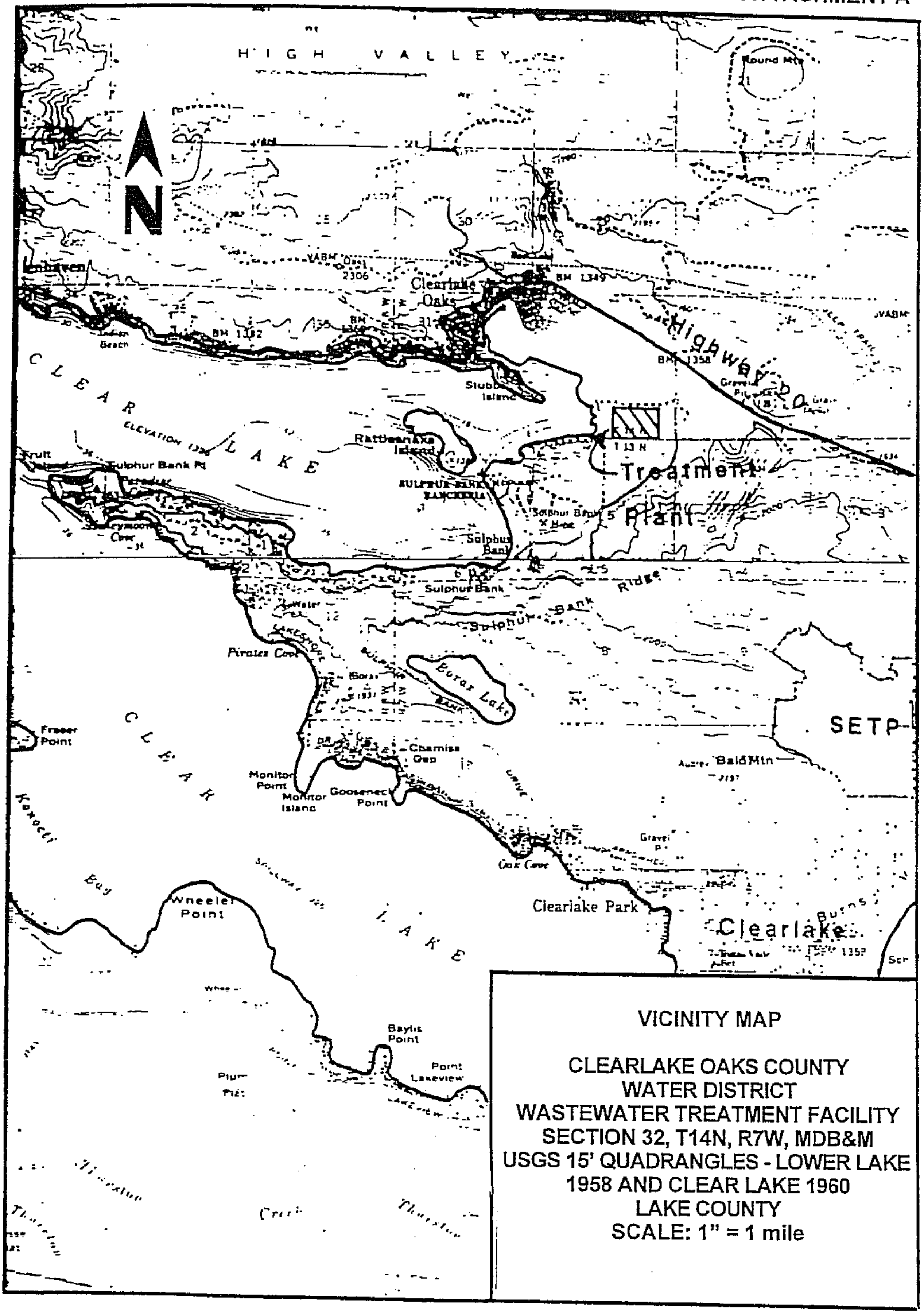
The Discharger shall implement the above monitoring program as of the date of this Order.

Ordered by: Gary M. Carlton
GARY M. CARLTON, Executive Officer

23 October 1998
(Date)

AMENDED
wac/lsb:clrlakeo.mrp

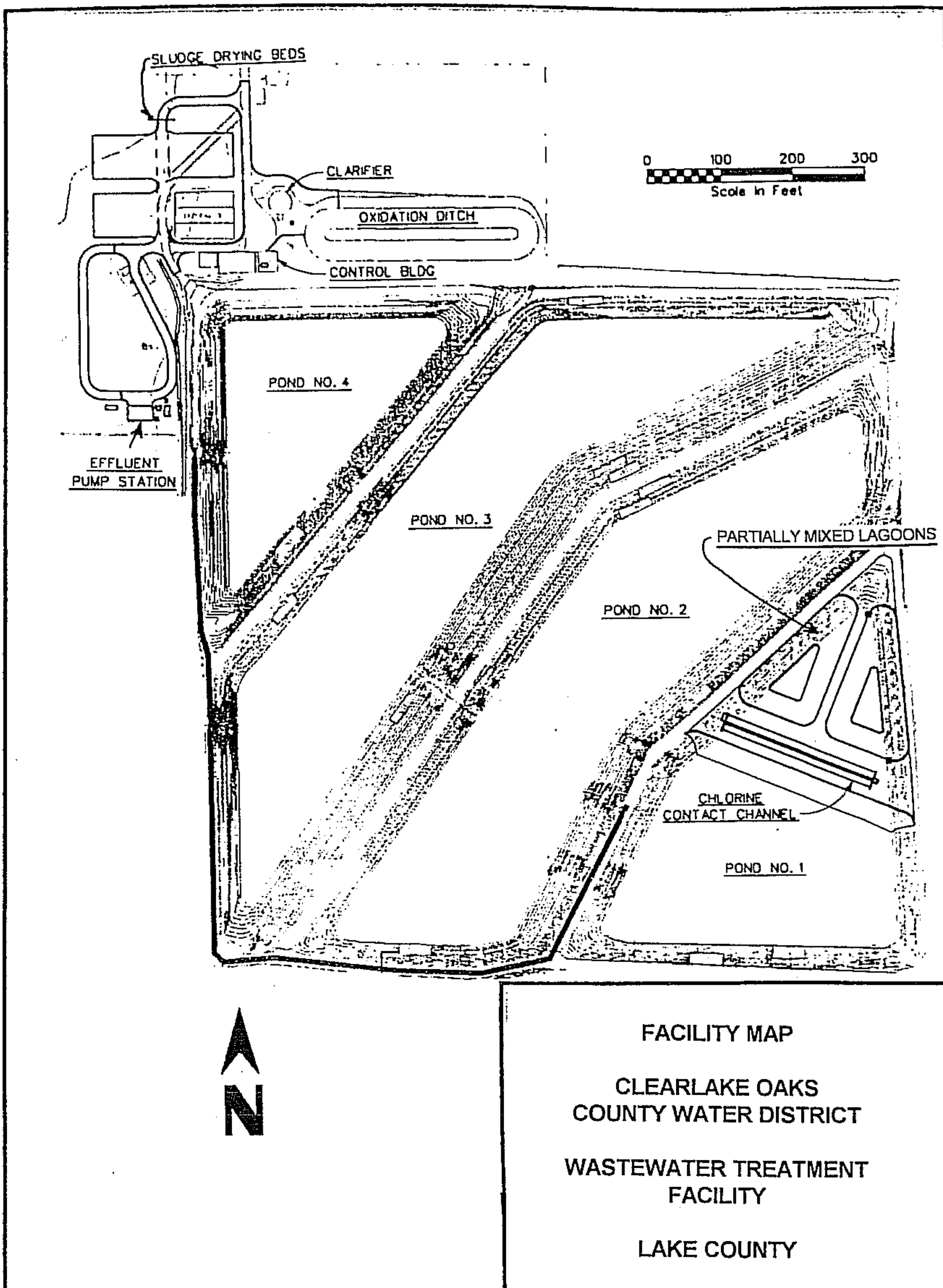
ATTACHMENT A



VICINITY MAP

CLEARLAKE OAKS COUNTY
WATER DISTRICT
WASTEWATER TREATMENT FACILITY
SECTION 32, T14N, R7W, MDB&M
USGS 15' QUADRANGLES - LOWER LAKE
1958 AND CLEAR LAKE 1960
LAKE COUNTY
SCALE: 1" = 1 mile

ATTACHMENT B

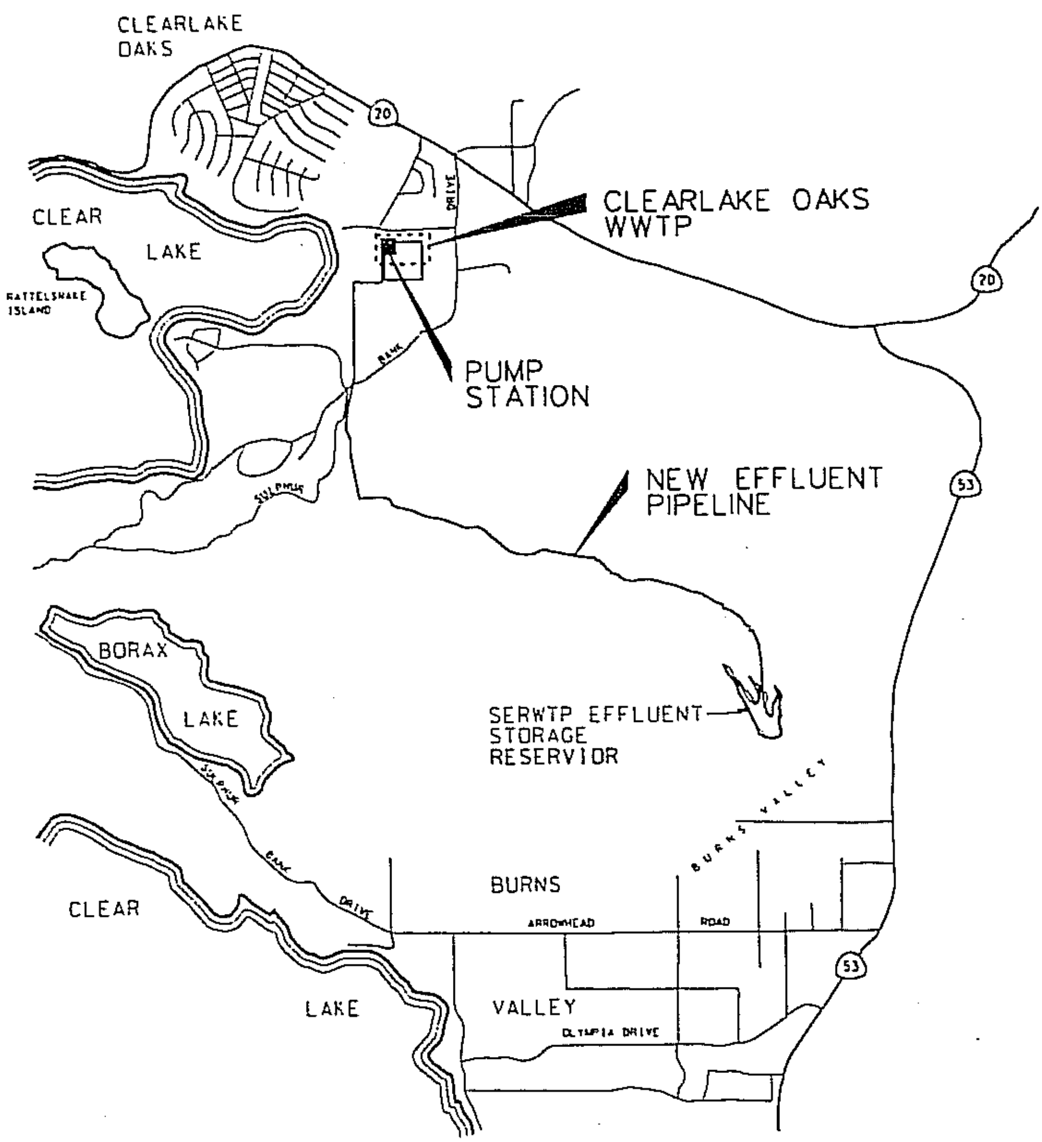


FACILITY MAP

CLEARLAKE OAKS
COUNTY WATER DISTRICT

WASTEWATER TREATMENT
FACILITY

LAKE COUNTY



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LOCATION MAP

CLEARLAKE OAKS COUNTY
WATER DISTRICT
WASTEWATER TREATMENT FACILITY
EFFLUENT PIPELINE PROJECT
LAKE COUNTY
NO SCALE

INFORMATION SHEET

CLEARLAKE OAKS COUNTY WATER DISTRICT
WASTEWATER TREATMENT PLANT
LAKE COUNTY

The Clearlake Oaks County Water District is on the eastern side of Clear Lake and services a community of approximately 2400. The District operates a secondary sewage treatment plant consisting of coarse screening, an oxidation ditch, clarifier, and evaporation/percolation ponds. The treatment plant has a design capacity of 0.5 mgd (average dry weather flow). Peak wet weather flow is 2.1 mgd.

The District has been plagued with inadequate disposal capacity when groundwater rises and reduces pond percolation. Lack of disposal capacity results in the District discharging to Clear Lake during wet winters in violation of waste discharge requirements and the Basin Plan discharge prohibition to Clear Lake. A Cease and Desist (C&D) Order was imposed in 1983 requiring the District to cease discharges to the lake.

The District responded to the C&D by proposing a seasonal discharge to a creek outside the immediate drainage basin. The Board adopted an NPDES Permit in 1988 which allowed the seasonal discharge of treated wastewater to Sweet Hollow Creek which is tributary to Cache Creek. Because of appeals and litigation by project opponents, the facilities for the Sweet Hollow Creek discharge were never built. The Board denied the five-year renewal of the NPDES Permit in 1993.

On 28 January 1994, the Board adopted Waste Discharge Requirements Order No. 94-028 which prescribed requirements for the treatment of approximately 0.43 million gallons per day of domestic wastewater in an oxidation ditch, and clarifier with discharge to disposal ponds. These requirements also included a groundwater assessment and monitoring program.

Order No. 94-028 is being revised to reflect treatment upgrades, collection system rehabilitations, emergency storage basin enhancements, and the installation of the effluent pipeline to the Southeast Regional Wastewater Treatment Facility.

The beneficial uses of Clear Lake are municipal, industrial, and agricultural supply; recreation; aesthetic enjoyment; navigation; groundwater recharge; fresh water replenishment; hydropower generation; and preservation and enhancement of fish, wildlife, and other aquatic resources.

The beneficial uses of underlying groundwater are domestic, agricultural, and industrial supply.

Surface water drainage is to Clear Lake.

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